# Project Initialization & Planning Phase: Project Proposal

Global Olympic Sports Performance Analysis (1992–2026)

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| **Date:** 15 March 2026 | **Team ID:** NOC-PERF-2026 |
| **Project Title:** Global Olympic Sports Performance Analysis | **Maximum Marks:** 3 Marks |

## Project Overview

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| **Objective** | Analyze 34 years of historical Olympic data (1992–2026) across 60 countries and 50 sports categories to evaluate per-athlete medal yield, regional investment efficiency, and host nation advantage. Deliver interactive Tableau dashboards and data stories to guide National Olympic Committees (NOCs), international grant agencies, and sports media in data-driven decision-making. |
| **Scope** | Scope includes ingesting historical athlete records and medal tallies (1992–2026), calculating per-athlete yield (Rowing: 6.916, Shooting: 6.731), analyzing regional efficiency (South America: 49.64%), benchmarking top powerhouses (USA: 735, China: 696), and tracking host dominance (Japan: 105). Excludes non-Olympic sporting events, direct athlete coaching programs, and real-time biometric tracking. |

## Problem Statement

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| **Description** | NOC strategists, sports funding agencies, and broadcast journalists struggle to extract actionable insights from vast, multi-decade historical Olympic datasets (1992–2026). Without normalized metrics—such as per-athlete medal return or investment efficiency ratios—countries misallocate funding to low-return sports, grant agencies overlook efficient but underfunded regions, and media networks lack data-driven host dominance narratives. |
| **Impact** | Solving this problem empowers NOCs to maximize medal haul per athlete, enables funding agencies to reallocate grants to high-potential underfunded regions (e.g., Slovakia, Georgia, Ethiopia, South America), and provides global broadcast networks with rich visual storytelling benchmarks. |

## Proposed Solution

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| **Approach** | Implement a 6-layer Business Intelligence pipeline using Python/Snowflake for ETL and data transformation, coupled with interactive Tableau dashboards. Compute calculated fields for per-athlete yield, regional efficiency ratios, and multi-decade host performance trends. |
| **Key Features** | • Horizontal Bar Charts displaying per-athlete medal return by sport category.<br>• Global Investment vs. Medal Performance Bubble Charts identifying overperforming nations.<br>• Regional Medal Efficiency Pie & Donut Charts.<br>• Multi-decade Host Dominance and Total Medal Line Charts (1992–2026). |

## Resource Requirements

| Resource Type | Description | Specification / Allocation |
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| **Hardware**<br>Computing Resources | High-performance workstation for dataset ETL processing and visualization rendering. | 8-core CPU (Intel i7/Apple M-series), 16 GB RAM |
| **Hardware**<br>Memory & Storage | Local staging storage for multi-game historical CSV exports and Tableau extracts. | 512 GB SSD, 16 GB System Memory |
| **Software**<br>Frameworks & Languages | Data extraction, numerical transformation, and SQL aggregation scripting. | Python 3.11, ANSI SQL |
| **Software**<br>Libraries & Tools | Data wrangling, analytical calculations, and interactive BI visual reporting. | pandas, numpy, Tableau Desktop / Prep, Snowflake |
| **Data** | Historical Olympic participation, medal outcomes, and regional sports funding records. | `athletes.csv` & `medal_results.csv` (1992–2026, 60 countries, 50 sports) |